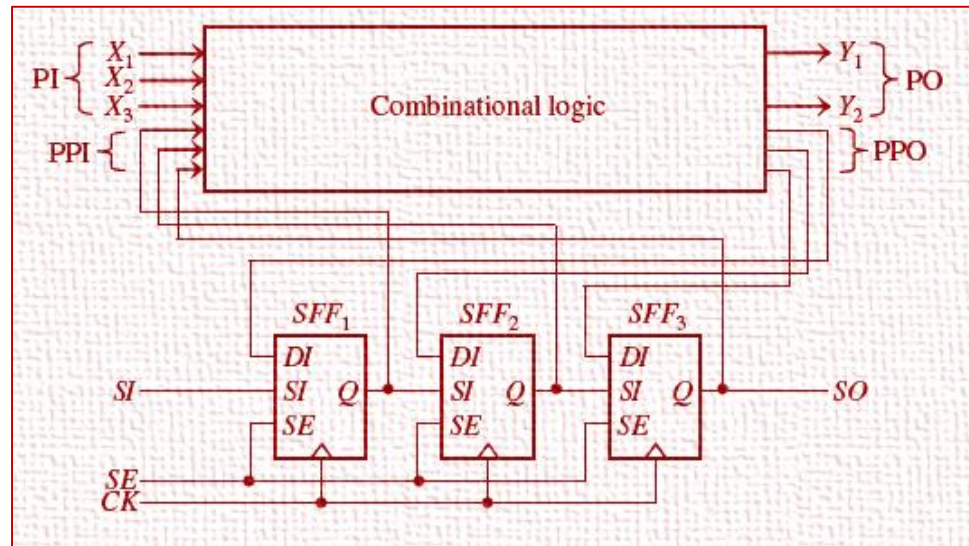


DFT - Part 1

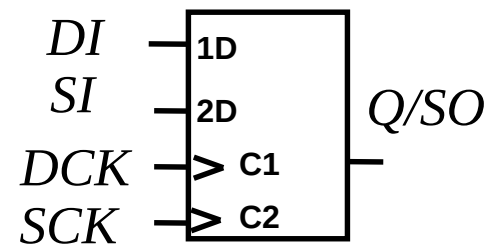
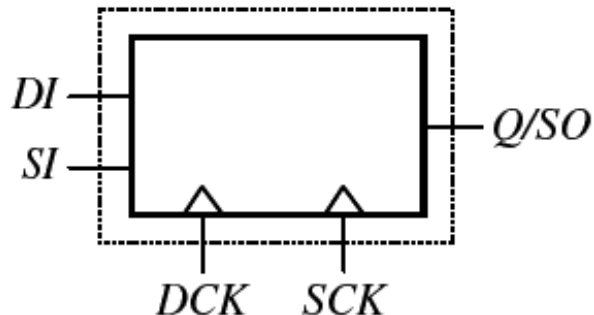
- Introduction
- Internal Scan
 - ◆ FF-based
 - * MUXed-D scan (1973, Stanford)
 - * Clocked scan (1968, 1975 NEC)
 - * Other scan
 - ◆ Latch-based
 - * LSSD (1977, IBM)
- Scan Design Flow
- Issues and Solutions
- Conclusion



Clocked Scan FF

[Kobayashi 68][Funatsu 75]

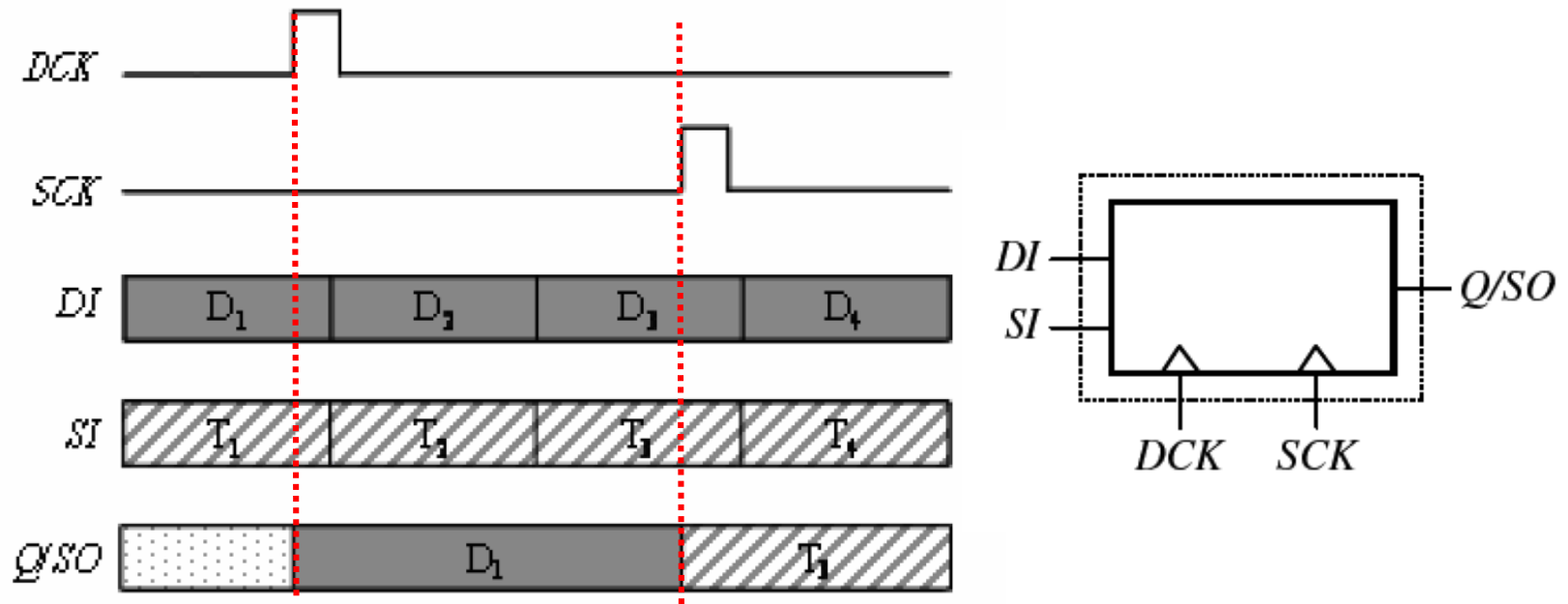
- 2 inputs
 - ♦ Data in (**DI**): from logic
 - ♦ Scan in (**SI**): from previous scan FF
- 2 independent clocks
 - ♦ Scan clock (**SCK**): capture **SI**
 - ♦ Data clock (**DCK**): capture **DI**
- Data output (**Q**) and Scan Output (**SO**) share same pin



IEEE symbol

Clocked Scan FF (2)

- $DI \rightarrow Q$ at positive DCK edge
- $SI \rightarrow Q$ at positive SCK edge

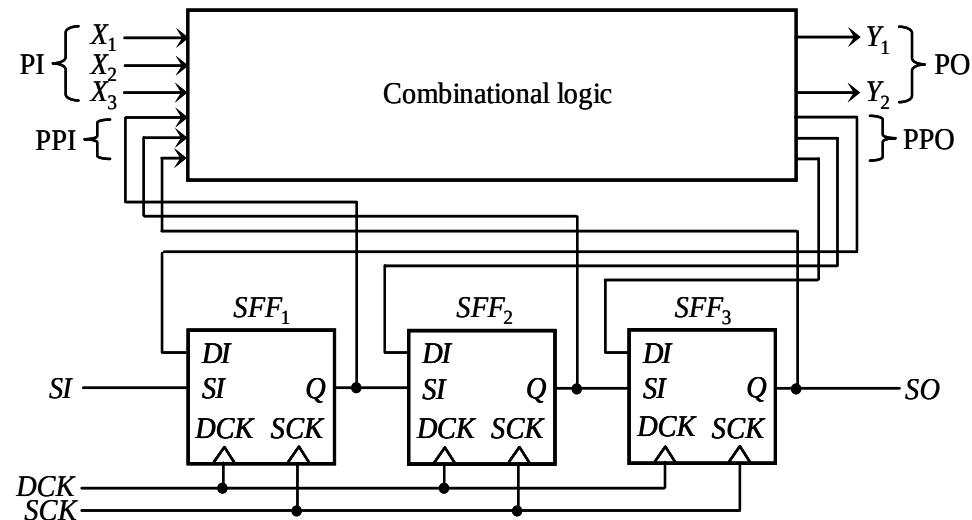
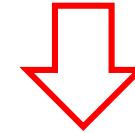
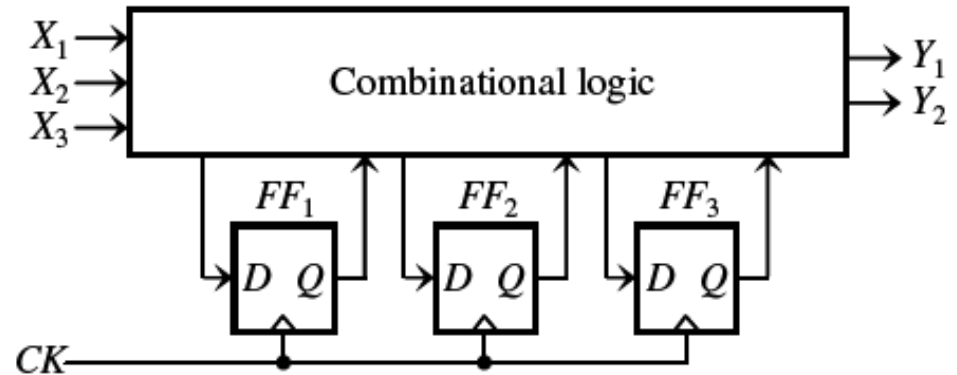


(WWW. Fig 2.11)

SCK/DCK Do NOT Overlap

Clocked Scan Architecture

- Original circuit
 - ♦ Single clock: CK
 - ♦ Regular D-FF
- After scan insertion
 - ♦ Two extra I/O pins
 - * SI, scan input
 - * SO, scan output
 - ♦ One extra clock
 - * SCK



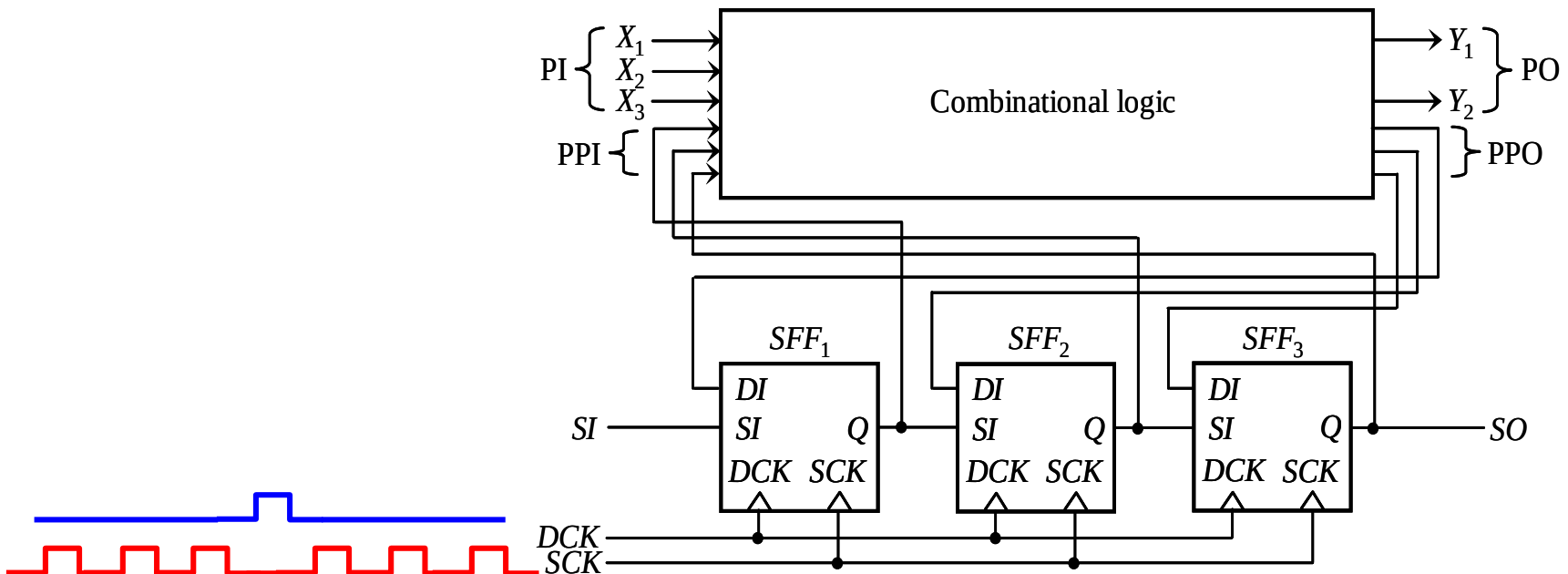
Clocked Scan Operation

- **Normal Mode**

- ◆ DCK, DCK ...

- **Test Mode**

- ◆ Shift: **SCK, SCK, SCK** (load scan chain)
 - ◆ Capture: **DCK**
 - ◆ Shift: **SCK, SCK, SCK** (unload scan chain)



Pros/Cons of Clocked Scan

- Advantage

 - ♦ **Faster than MUXed-D scan**

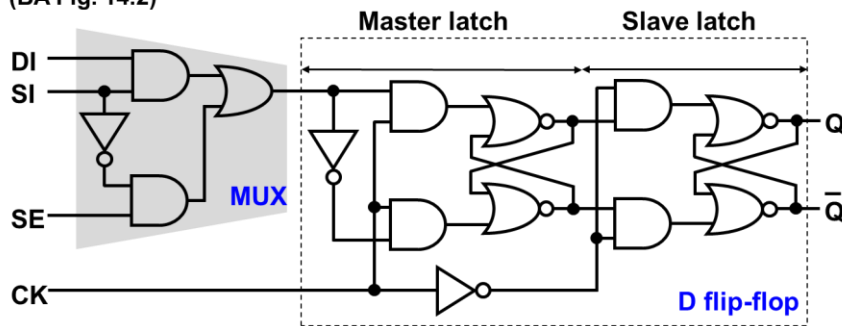
 - * Less delay overhead

- Disadvantage

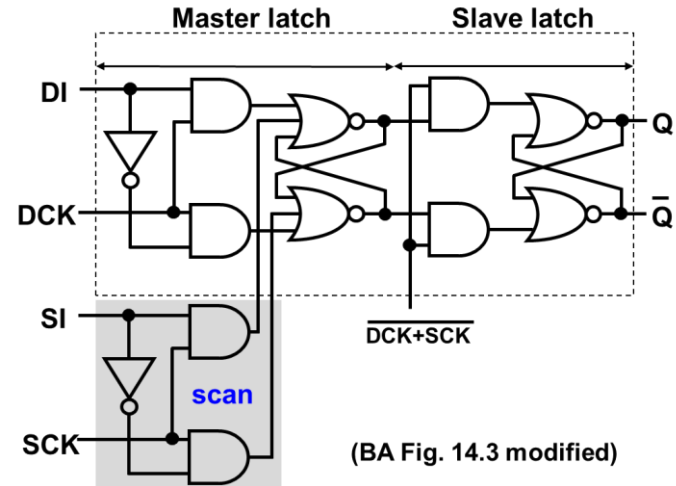
 - ♦ **Larger routing overhead** than MUXed-D scan

 - * Needs one extra clock distribution (SCK)

(BA Fig. 14.2)



MUXed-D scan



(BA Fig. 14.3 modified)

Clocked scan

CS Faster but Larger

Quiz

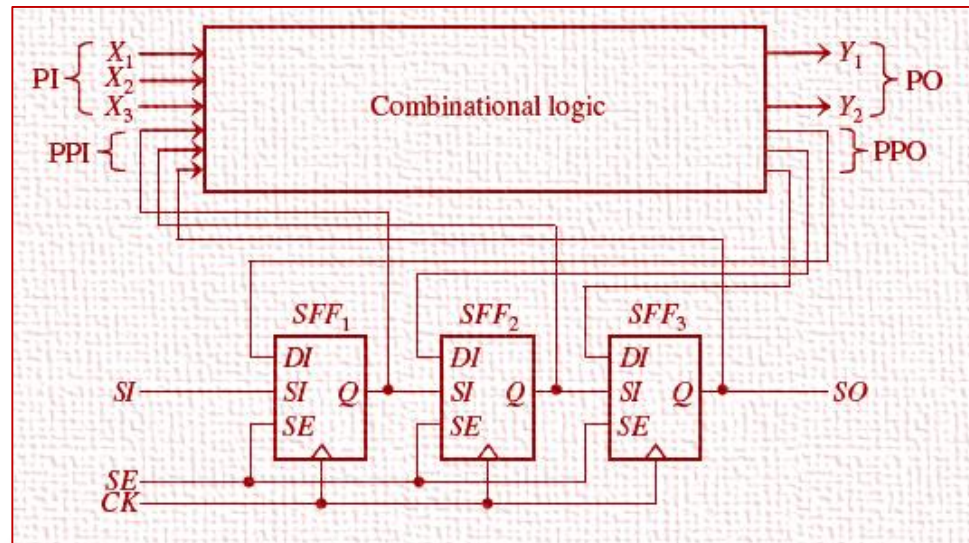
Q: Which of the following is NOT true about clocked scan?

- A. Clocked scan is better than MUXed-D scan**
- B. Clocked scan has two clocks**
- C. Clocked scan is faster than MUXed-D scan**
- D. Clocked scan is useful for high speed circuits**

ANS:

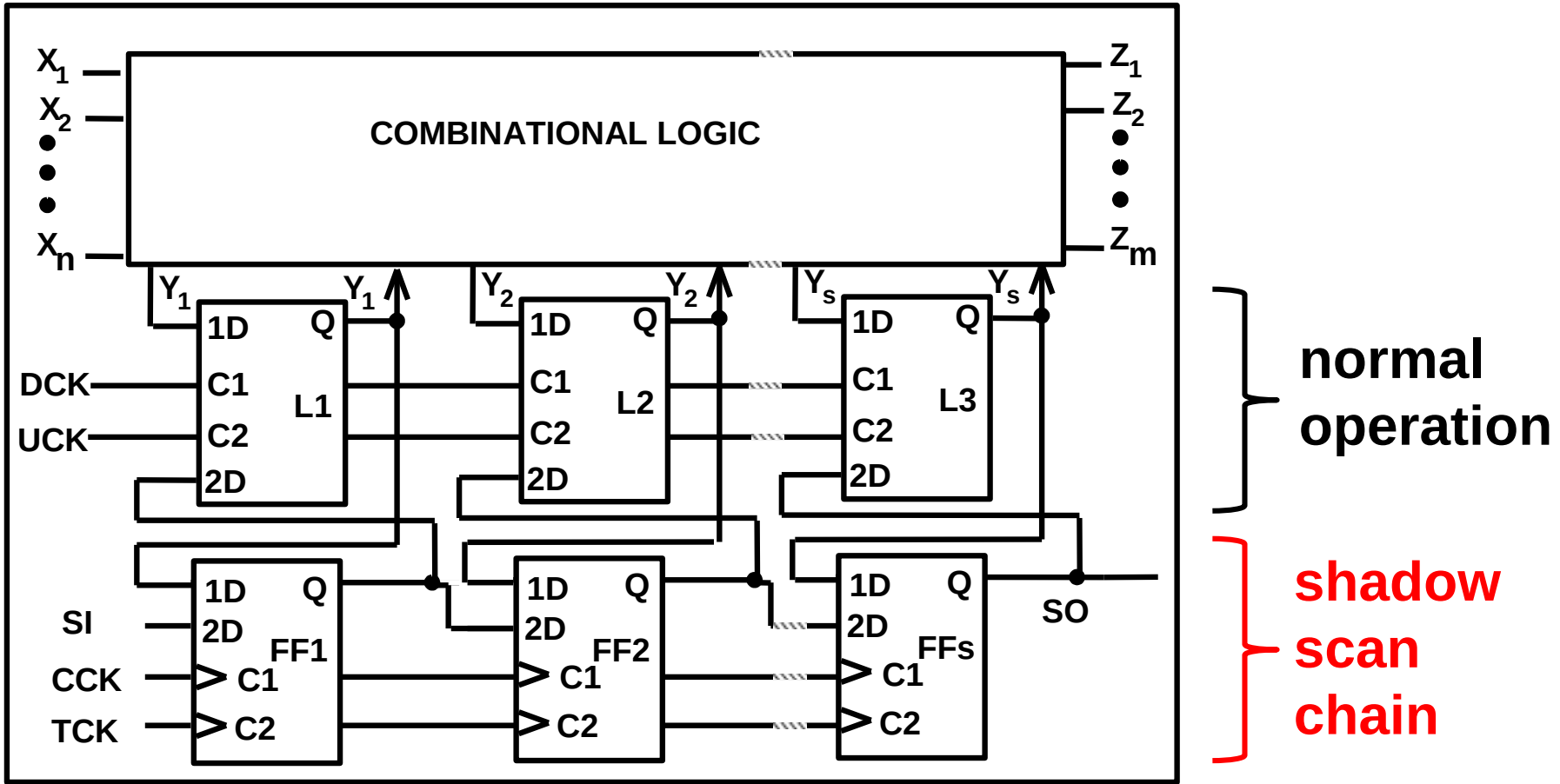
DFT - Part 1

- Introduction
- Internal Scan
 - ◆ FF-based
 - * MUXed-D scan (1973, Stanford)
 - * Clocked scan (1968, 1975 NEC)
 - * Other scan
 - Shadow scan chain
 - Random access scan, scan tree... (not in lecture)
 - ◆ Latch-based
 - * LSSD (1977, IBM)
- Scan Design Flow
- Issues and Solutions
- Conclusion



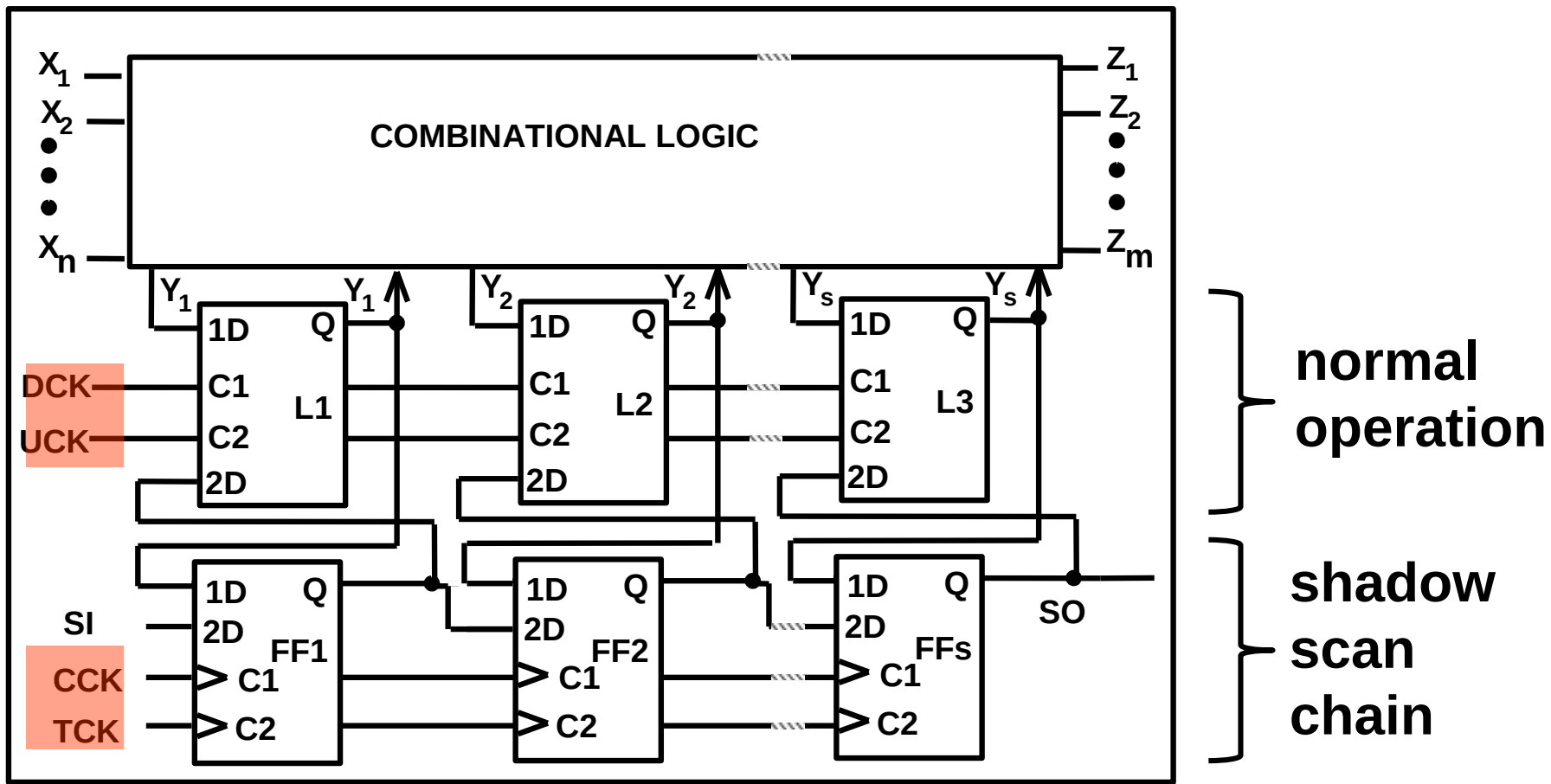
Shadow Scan Chain

- **Design for Debug (DfD)**. More expensive than DfT
 - ◆ Duplicate FF!
- Purpose: **Allow normal operation** during scan chain shifting



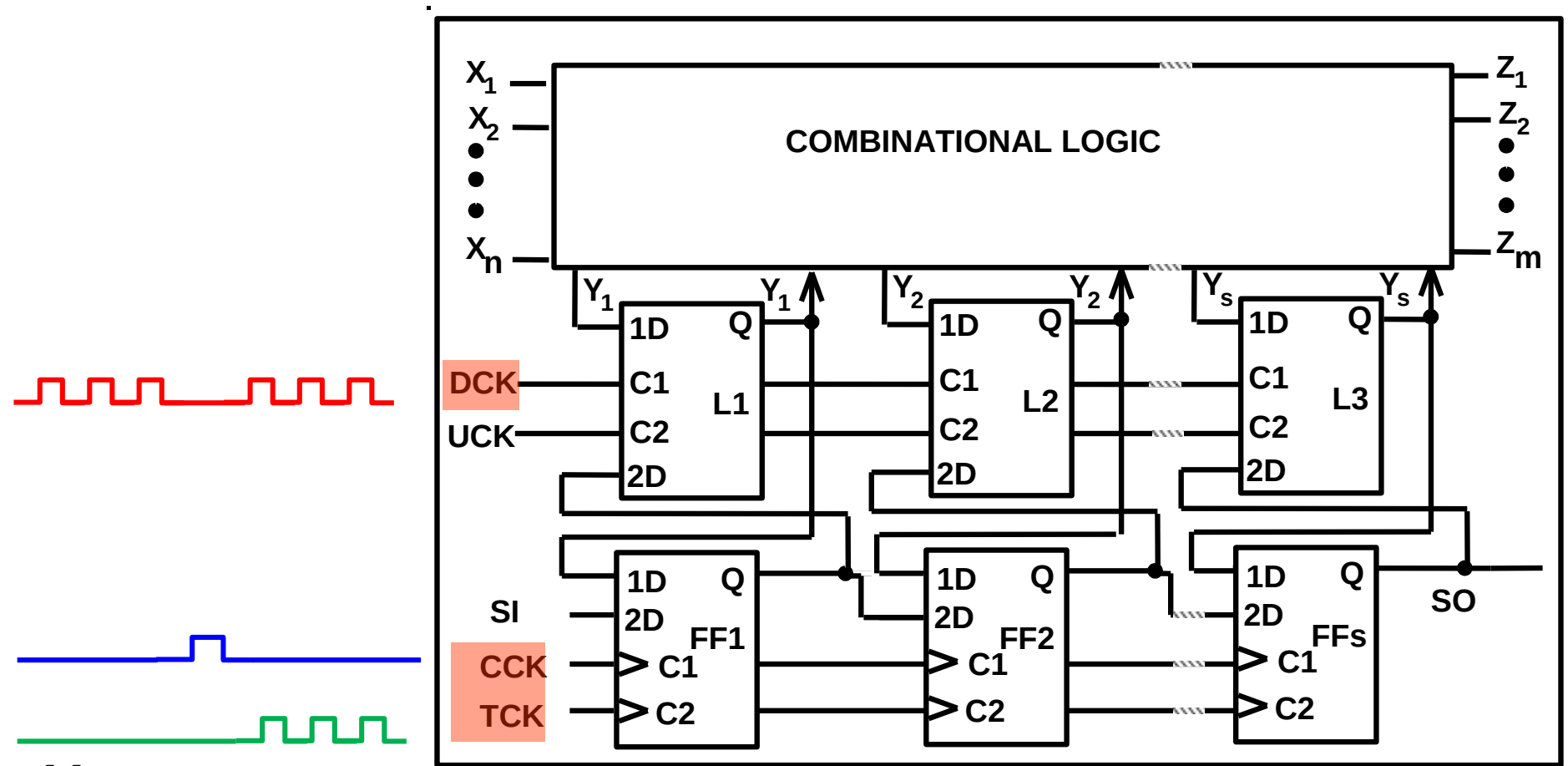
Shadow Scan Chain (2)

- Four clocks!
 - ♦ Normal: **DCK**
 - ♦ Capture: **CCK**. Shift: **TCK**. Update: **UCK**



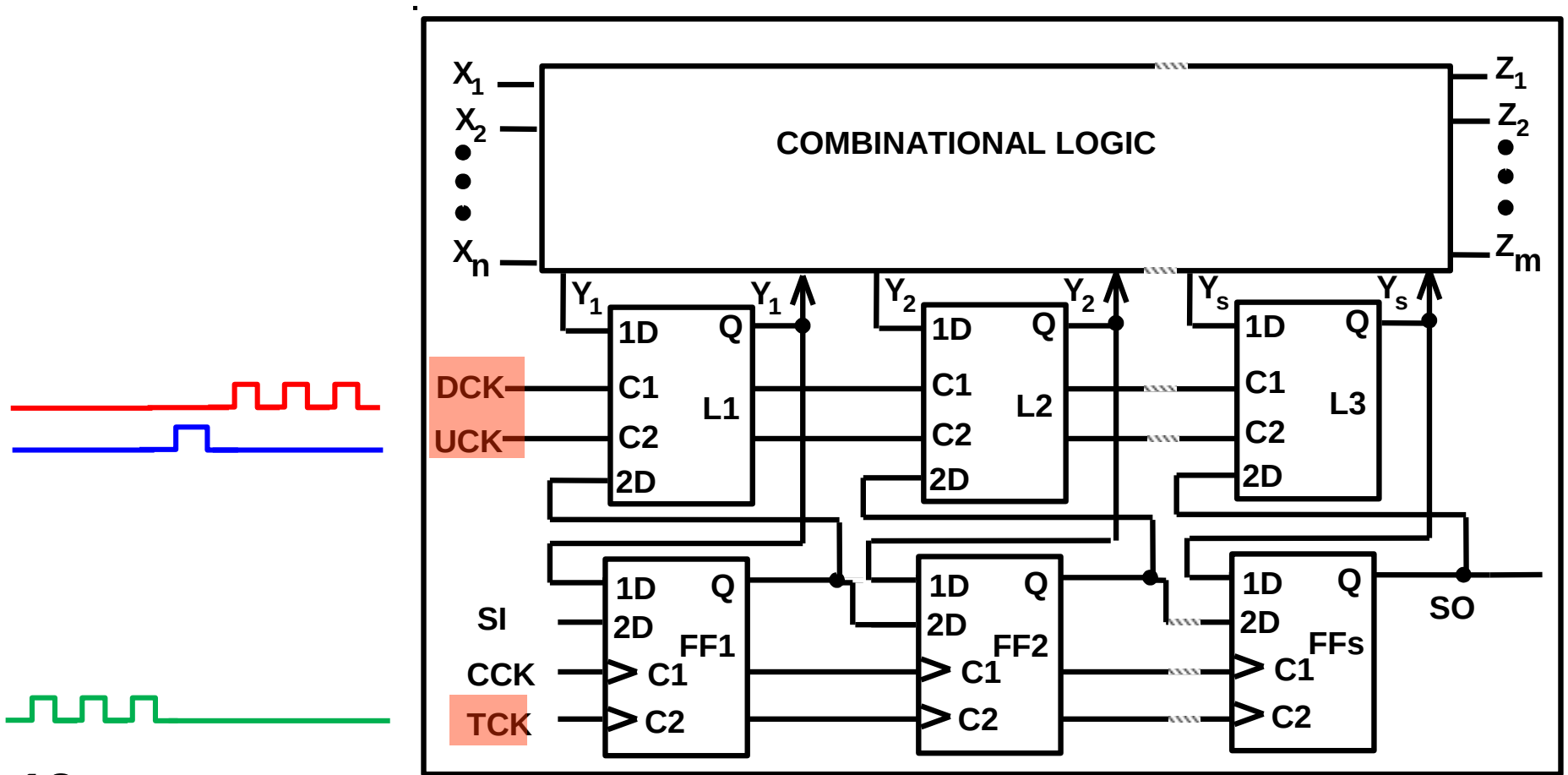
Observe FF Contents

- Normal operation: **DCK, DCK ...**
 - ♦ Capture: **CCK**
 - ♦ Shift out: **TCK, TCK, TCK**. Normal operation: **DCK, DCK**



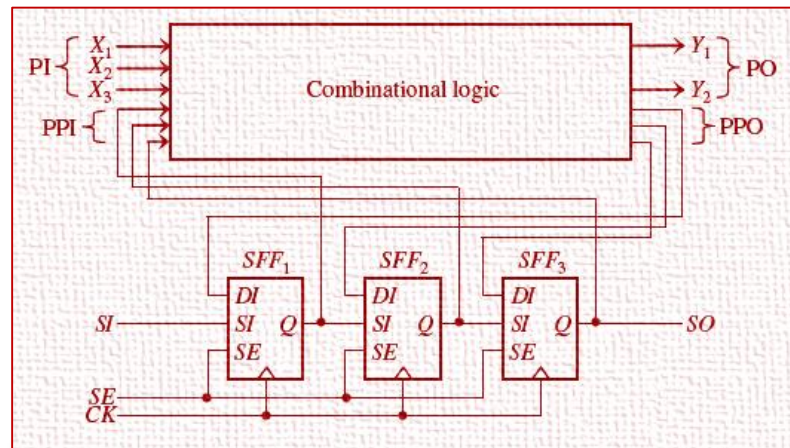
Control FF Contents

- Shift in: **TCK**, **TCK**, **TCK**
 - ◆ Update: **UCK**
 - ◆ Begin normal mode: **DCK**, **DCK** ...



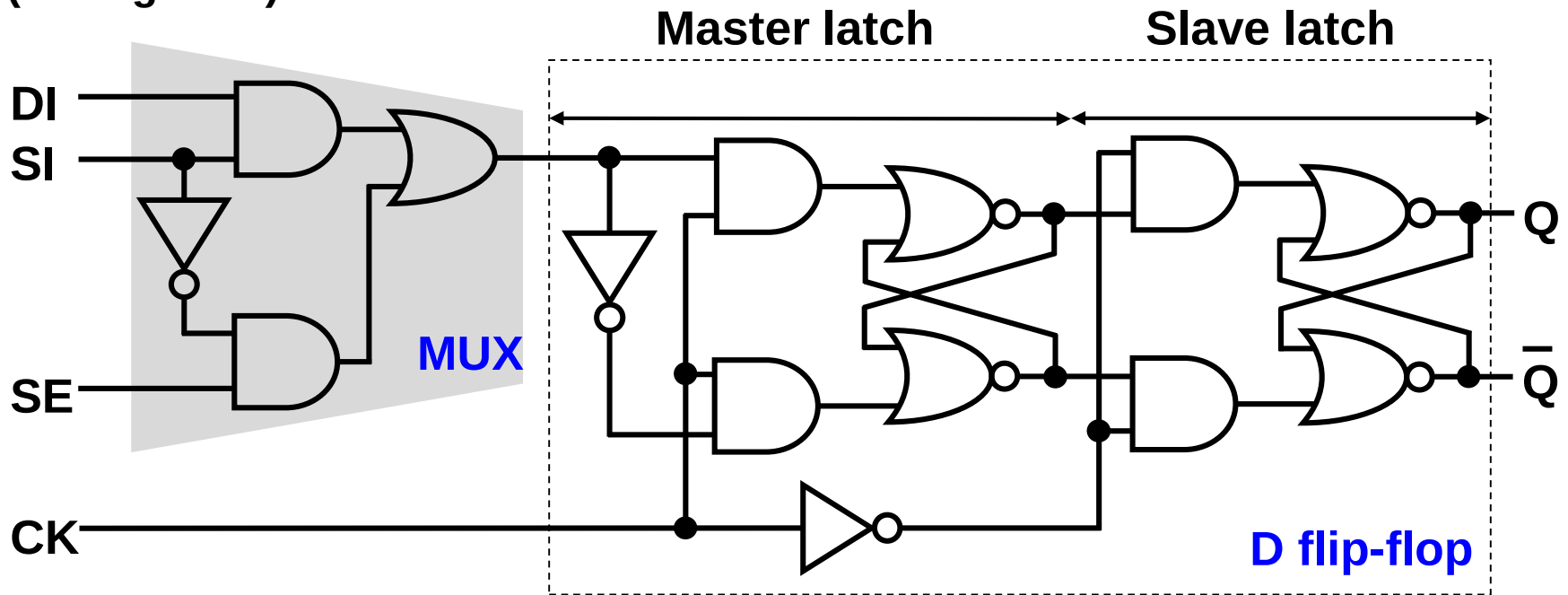
Summary

- Clocked scan : Two clocks
 - 😊 **Faster** than MUXed-D scan
 - 😞 **Larger** than MUXed-D scan
 - * Useful for high speed circuits
- Shadow scan :
 - ◆ **Control** and **observe** contents of FF during normal operation
 - 😊 Useful **Design for Debug (DfD)**
 - 😞 **Hugh area overhead !**
 - * Used in very expensive circuits



APPENDIX: Schematic of MUXed-D Scan

(BA Fig. 14.2)



APPENDIX: Schematic of Clocked Scan

